Faculty of Technology and Engineering

Chandubhai S. Patel Institute of Technology (CSPIT) Department of Computer Science & Engineering

Date: 25/10/2026

Laboratory Manual

Academic Year	:	2026-27	Semester	:	3rd
Course code	:	ITUE203	Course name	:	WEB DEVELOPMENT FRAMEWORKS

Practical – 14

PROBLEM STATEMENT :

Develop an admin dashboard that displays user statistics, student count, event count and latest registrations. Restrict access to admin users only and show dynamic menu options based on role.

OBJECTIVE:

To design and implement a secure, **role-based administrative dashboard** that provides real-time data visualization and dynamic navigation management. The system aims to restrict data access exclusively to authorized **admin users** while dynamically rendering menu options based on user roles, ensuring secure and optimized data management.

CODE:

db.php

```php
<?php
// includes/db.php : PDO database connection
$DB_HOST = 'localhost';
$DB_NAME = 'admin_dashboard';
$DB_USER = 'root';    // XAMPP default
$DB_PASS = '';        // XAMPP default (empty)

try {
    $pdo = new PDO(
        "mysql:host=$DB_HOST;dbname=$DB_NAME;charset=utf8mb4",
        $DB_USER,
        $DB_PASS,
        [
            PDO::ATTR_ERRMODE            => PDO::ERRMODE_EXCEPTION,
            PDO::ATTR_DEFAULT_FETCH_MODE => PDO::FETCH_ASSOC,
        ]
    );
} catch (PDOException $e) {
    die('Database connection failed. Check includes/db.php and import
schema.sql.');
}
```

dashboard.php

```php
<?php
// dashboard.php : ADMIN ONLY. Stats + latest registrations
require_once __DIR__ . '/includes/auth.php';
require_role(['admin']);              // RBAC check -> 403 for everyone
else

/* ---------- Dashboard queries ---------- */
$totalUsers    = (int)$pdo->query('SELECT COUNT(*) FROM users')-
>fetchColumn();
$totalStudents = (int)$pdo->query('SELECT COUNT(*) FROM students')-
>fetchColumn();
$totalEvents   = (int)$pdo->query('SELECT COUNT(*) FROM events')-
>fetchColumn();
$totalRegs     = (int)$pdo->query('SELECT COUNT(*) FROM registrations')-
>fetchColumn();

$roleStats = $pdo->query(
    'SELECT r.name, COUNT(u.id) AS total
     FROM roles r LEFT JOIN users u ON u.role_id = r.id
     GROUP BY r.id, r.name'
)->fetchAll();
```

```
$latest = $pdo->query(
    'SELECT s.name AS student, s.enrollment_no, e.title AS event,
g.registered_at
     FROM registrations g
     JOIN students s ON s.id = g.student_id
     JOIN events   e ON e.id = g.event_id
     ORDER BY g.registered_at DESC
     LIMIT 5'
)->fetchAll();

log_action('VIEW_DASHBOARD', 'Admin opened dashboard');

$pageTitle = 'Dashboard';
require __DIR__ . '/includes/header.php';

$cards = [
    ['Total Users',   $totalUsers,    'bi-person-badge',   'bg-primary'],
    ['Students',      $totalStudents, 'bi-mortarboard',    'bg-success'],
    ['Events',        $totalEvents,   'bi-calendar-event', 'bg-warning
text-dark'],
    ['Registrations', $totalRegs,     'bi-clipboard-check','bg-danger'],
];
?>
<h3 class="mb-4">Welcome, <?= e(current_user()['name']) ?> 👋</h3>

<div class="row g-3 mb-4">
  <?php foreach ($cards as [$label, $value, $icon, $bg]): ?>
    <div class="col-6 col-lg-3">
      <div class="card stat-card <?= $bg ?> shadow-sm">
        <div class="card-body d-flex justify-content-between align-items-
center">
          <div>
            <div class="small text-uppercase"><?= $label ?></div>
            <div class="display-6 fw-bold"><?= $value ?></div>
          </div>
          <i class="bi <?= $icon ?>"></i>
        </div>
      </div>
    </div>
  <?php endforeach; ?>
</div>

<div class="row g-3">
  <div class="col-lg-8">
    <div class="card shadow-sm border-0">
      <div class="card-header bg-white fw-semibold">Latest
Registrations</div>
      <div class="table-responsive">
        <table class="table table-hover mb-0 align-middle">
          <thead><tr><th>#</th><th>Student</th><th>Enrollment</th><th>Even
t</th><th>Registered At</th></tr></thead>
          <tbody>
```

```
          <?php foreach ($latest as $i => $r): ?>
            <tr>
              <td><?= $i + 1 ?></td>
              <td><?= e($r['student']) ?></td>
              <td><?= e($r['enrollment_no']) ?></td>
              <td><?= e($r['event']) ?></td>
              <td><?= e($r['registered_at']) ?></td>
            </tr>
          <?php endforeach; ?>
          <?php if (!$latest): ?><tr><td colspan="5" class="text-center
text-muted">No registrations yet.</td></tr><?php endif; ?>
          </tbody>
        </table>
      </div>
    </div>
  </div>

  <div class="col-lg-4">
    <div class="card shadow-sm border-0">
      <div class="card-header bg-white fw-semibold">Users by Role</div>
      <ul class="list-group list-group-flush">
        <?php foreach ($roleStats as $r): ?>
          <li class="list-group-item d-flex justify-content-between text-
capitalize">
            <?= e($r['name']) ?>
            <span class="badge bg-dark rounded-pill"><?= (int)$r['total']
?></span>
          </li>
        <?php endforeach; ?>
      </ul>
    </div>
  </div>
</div>
<?php require __DIR__ . '/includes/footer.php'; ?>
```

schema.sql

```
-- schema.sql : Admin Dashboard practical
CREATE DATABASE IF NOT EXISTS admin_dashboard CHARACTER SET utf8mb4;
USE admin_dashboard;

DROP TABLE IF EXISTS audit_logs;
DROP TABLE IF EXISTS registrations;
DROP TABLE IF EXISTS events;
DROP TABLE IF EXISTS students;
DROP TABLE IF EXISTS users;
DROP TABLE IF EXISTS roles;

CREATE TABLE roles (
```

```sql
  id   TINYINT PRIMARY KEY AUTO_INCREMENT,
  name VARCHAR(30) NOT NULL UNIQUE
);

CREATE TABLE users (
  id            INT PRIMARY KEY AUTO_INCREMENT,
  name          VARCHAR(100) NOT NULL,
  email         VARCHAR(150) NOT NULL UNIQUE,
  password_hash VARCHAR(255) NOT NULL,
  role_id       TINYINT NOT NULL,
  created_at    TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (role_id) REFERENCES roles(id)
);

CREATE TABLE students (
  id            INT PRIMARY KEY AUTO_INCREMENT,
  enrollment_no VARCHAR(20) NOT NULL UNIQUE,
  name          VARCHAR(100) NOT NULL,
  department    VARCHAR(60) NOT NULL,
  created_at    TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE events (
  id         INT PRIMARY KEY AUTO_INCREMENT,
  title      VARCHAR(150) NOT NULL,
  venue      VARCHAR(100) NOT NULL,
  event_date DATE NOT NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE registrations (
  id            INT PRIMARY KEY AUTO_INCREMENT,
  student_id    INT NOT NULL,
  event_id      INT NOT NULL,
  registered_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (student_id) REFERENCES students(id) ON DELETE CASCADE,
  FOREIGN KEY (event_id)   REFERENCES events(id)   ON DELETE CASCADE
);

CREATE TABLE audit_logs (
  id         INT PRIMARY KEY AUTO_INCREMENT,
  user_id    INT NULL,
  action     VARCHAR(50) NOT NULL,
  details    VARCHAR(255) NULL,
  ip_address VARCHAR(45) NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE SET NULL
);

-- ---------- Seed data ----------
INSERT INTO roles (name) VALUES ('admin'), ('faculty'), ('student');
```

```
-- All three demo users have the password:  password
INSERT INTO users (name, email, password_hash, role_id) VALUES
('Admin
User',   'admin@demo.com',   '$2y$10$92IXUNpkjO0rOQ5byMi.Ye4oKoEa3Ro9llC/.
og/at2.uheWG/igi', 1),
('Faculty User', 'faculty@demo.com',
'$2y$10$92IXUNpkjO0rOQ5byMi.Ye4oKoEa3Ro9llC/.og/at2.uheWG/igi', 2),
('Student User', 'student@demo.com',
'$2y$10$92IXUNpkjO0rOQ5byMi.Ye4oKoEa3Ro9llC/.og/at2.uheWG/igi', 3);

INSERT INTO students (enrollment_no, name, department) VALUES
('23IT001', 'Aarav Patel',  'IT'),
('23IT002', 'Diya Shah',    'IT'),
('23CE010', 'Rohan Mehta',  'CE'),
('23CS021', 'Isha Desai',   'CSE'),
('23IT033', 'Kunal Joshi',  'IT');

INSERT INTO events (title, venue, event_date) VALUES
('Hackathon 2026',       'Main Auditorium', '2026-11-10'),
('Web Dev Workshop',     'Lab 301',         '2026-11-18'),
('Tech Fest Quiz',       'Seminar Hall',    '2026-12-02');

INSERT INTO registrations (student_id, event_id, registered_at) VALUES
(1, 1, NOW() - INTERVAL 5 DAY),
(2, 1, NOW() - INTERVAL 4 DAY),
(3, 2, NOW() - INTERVAL 3 DAY),
(4, 2, NOW() - INTERVAL 2 DAY),
(5, 3, NOW() - INTERVAL 1 DAY),
(1, 3, NOW());
```

auth.php

```
<?php
// includes/auth.php : session, RBAC check, audit logging, dynamic menu
session_start();
require_once __DIR__ . '/db.php';

/* ---------- Menu definition: [label, link, icon, roles allowed to see
it] ---------- */
const MENU = [
    ['Dashboard',    'dashboard.php',  'bi-speedometer2',  ['admin']],
    ['Students',     'students.php',   'bi-people',        ['admin',
'faculty']],
    ['Events',       'events.php',     'bi-calendar-event',['admin',
'faculty', 'student']],
    ['Manage Users', 'users.php',      'bi-person-gear',   ['admin']],
    ['Audit Logs',   'audit_logs.php', 'bi-journal-text',  ['admin']],
];

function current_user(): ?array {
    return $_SESSION['user'] ?? null;
}
```

```php
function require_login(): void {
    if (!current_user()) {
        header('Location: login.php');
        exit;
    }
}

/* ---------- RBAC check: only the listed roles may continue ---------- */
function require_role(array $roles): void {
    require_login();
    if (!in_array(current_user()['role'], $roles, true)) {
        log_action('ACCESS_DENIED', 'Tried to open ' .
basename($_SERVER['PHP_SELF']));
        http_response_code(403);
        require __DIR__ . '/../403.php';
        exit;
    }
}

/* ---------- Audit log writer ---------- */
function log_action(string $action, string $details = ''): void {
    global $pdo;
    $uid = current_user()['id'] ?? null;
    $stmt = $pdo->prepare(
        'INSERT INTO audit_logs (user_id, action, details, ip_address)
VALUES (?, ?, ?, ?)'
    );
    $stmt->execute([$uid, $action, $details, $_SERVER['REMOTE_ADDR'] ??
null]);
}

/* ---------- Menu items visible for the logged-in role ---------- */
function menu_for_role(string $role): array {
    return array_values(array_filter(MENU, fn($m) => in_array($role,
$m[3], true)));
}

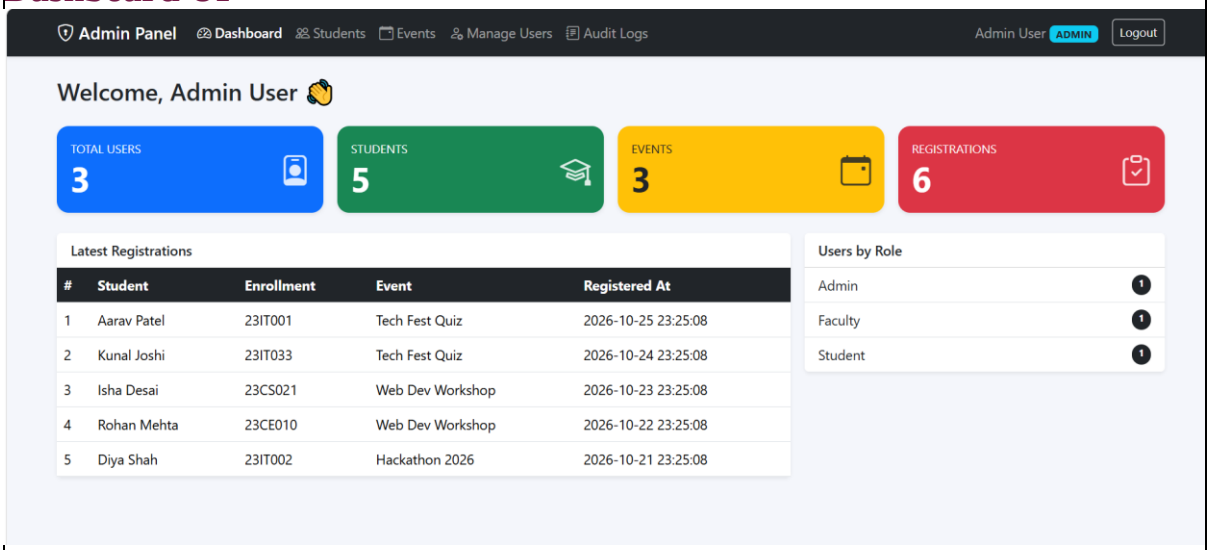
/* ---------- CSRF helpers ---------- */
function csrf_token(): string {
    if (empty($_SESSION['csrf'])) {
        $_SESSION['csrf'] = bin2hex(random_bytes(16));
    }
    return $_SESSION['csrf'];
}

function csrf_check(): void {
    if (!hash_equals($_SESSION['csrf'] ?? '', $_POST['csrf'] ?? '')) {
        http_response_code(400);
        die('Invalid CSRF token.');
    }
}
```

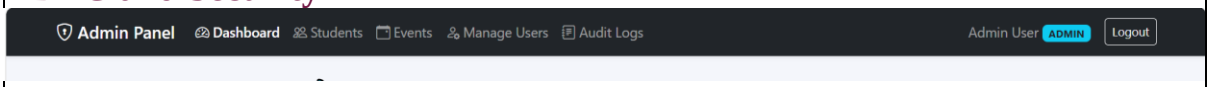
```php
function e(?string $s): string {
    return htmlspecialchars((string)$s, ENT_QUOTES, 'UTF-8');
}
```

SCREENSHOT:

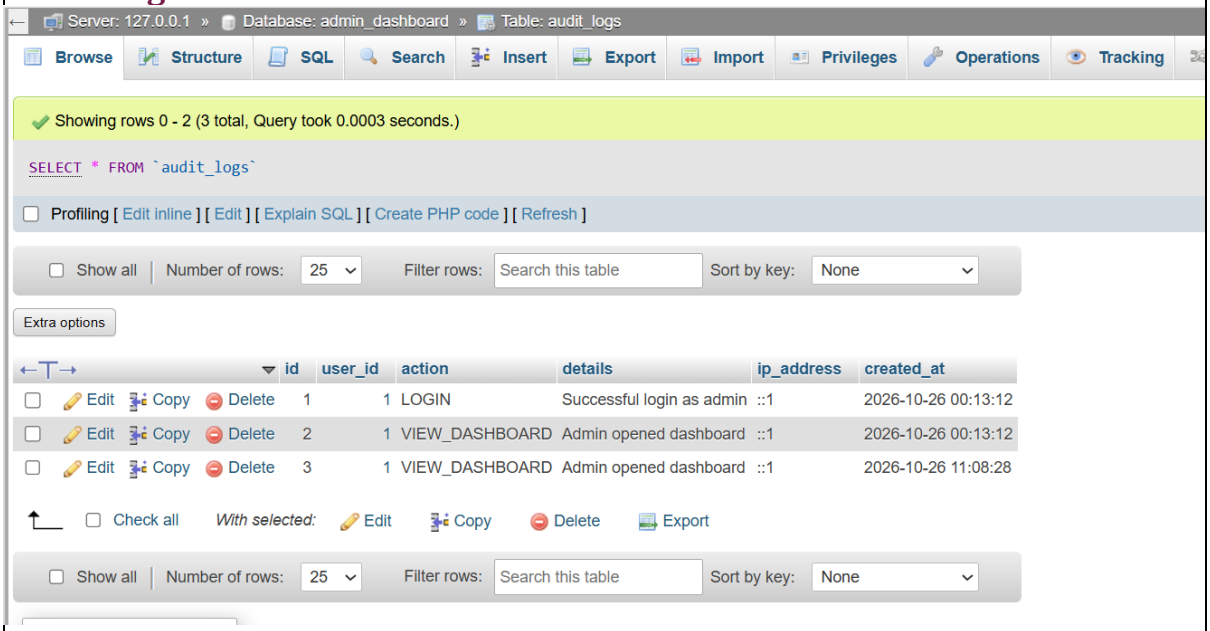
Dashboard UI

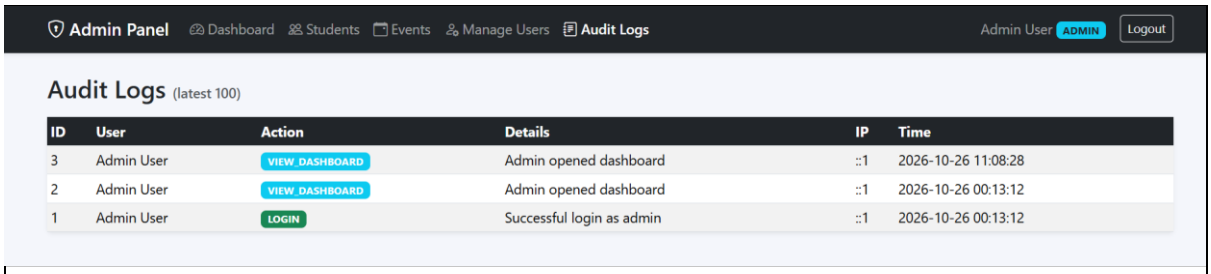


RBAC and Security



Audit Log Proof





LEARNING OUTCOME:

Students will build a role-based admin dashboard with basic monitoring and traceability.